MILITARY SPECIFICATION

ELECTRICAL SYSTEMS, AIRCRAFT, DESIGN AND INSTALLATION OF, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-E-25499C, dated 23 March 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1, paragraph 2.1: Under Military Specifications delete MIL-E-25704 and title thereto and add:

"MIL-G-21480	Generator System, 400 Hertz Alternating Current, Aircraft General Specification For
MIL-E-23001	Electric Generating System, Variable Speed Constant Frequency, Aircraft, General Specification For
MIL-D-83029	Drive Assembly, Constant Speed, General Specification For
MIL-B-83769	Battery, Storage, 24 Volt, Lead Acid"

Page 3: Add:

"3.1.7 Dielectric strength. The electrical insulation of equipment, except voltage-sensitive components such as semiconductors and capacitors, shall be capable of withstanding, at commercial frequencies, the application of voltage equal to 1000 volts rms plus twice the rated working voltage."

Page 6, paragraphs 3.2.2.4.2 and 3.2.2.4.2.1: Delete and substitute:

"3.2.2.4.2 Battery system. The battery system shall consist of a battery, provisions for battery charging, and a battery relay."

"3.2.2.4.2.1 Battery. The battery shall be of a type and rating approved by the procuring activity. The specified ampere-hour capacity, maximum discharge current, minimum voltage, and environmental capabilities of the selected battery shall meet the requirements of the application."

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"3.2.2.4.2.2 Battery charging. Battery charging shall be controlled to enable the battery to become fully-charged without excessive gassing or heating throughout the entire range of battery operating temperatures. Nickel-cadmium charging systems shall sense battery cell temperature and shall provide temperature compensation for charging rate and charge termination point. Charging of nickel-cadmium batteries shall be prevented for cell temperatures in excess of 60°C (140°F)."

"3.2.2.4.2.3 Battery relay. A relay shall be installed in each battery circuit in accordance with MS 24439 to enable the battery to be isolated from the normal system. The relay shall be located as close as practicable to the battery. The battery switch shall be installed on the pilot's or engineer's control panel for the electrical system. Any circuits which require battery power with the battery switch off shall be connected directly to the battery through suitable fuses or circuit breakers."

Page 6, paragraph 3.2.2.5.1: Delete and substitute:

"3.2.2.5.1 A-C generator systems. A-C generator systems shall be in accordance with MIL-G-21480 or MIL-E-23001. For Air Force use, constant speed drives shall be in accordance with MIL-D-83029."

Page 6, paragraph 3.2.2.5.2: Delete and substitute:

"3.2.2.5.2 Inverter power system. Inverters shall be in accordance with MIL-I-7032 or MIL-I-27273."

Page 9, paragraph 4.2: Delete and substitute:

"4.2 Tests. Electrical system mock-up tests shall be conducted in accordance with MIL-E-25500; flight tests shall be in accordance with MIL-F-25381; environmental testing on electrical equipment shall be in accordance with MIL-STD-810; and electromagnetic interference tests shall be in accordance with MIL-STD-462."

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